

Setup Guide: Color Sorter

Time Allocation: 90 Minutes

Challenge Overview

The robot must identify and push three different colored blocks into their corresponding colored drop zones.

Environment & Arena Setup

- **Arena Surface:** Use a flat, smooth, and non-reflective white or light-gray baseboard or table (approximately 4x4 feet) to serve as the sorting arena.
- **Drop Zones:** Mark three distinct drop zones in designated corners or edges of the arena using colored tape or construction paper corresponding to the three block colors (e.g., Red, Blue, Green).
- **Starting Box:** Tape a designated 'Start Zone' near the center of the arena where the robot must begin every run.
- **Block Placement:** Provide a mixed stash of nine total blocks (three of each color). Instructors should place one block of each color in a randomized starting array in front of the robot, or have students draw blocks blindly for each trial to test adaptability.